

System Analysis & Design: FurniZone Project

1. Problem Statement & Objectives

- **Problem Statement:** The traditional furniture shopping experience often lacks variety and convenience. FurniZone addresses this by providing a digital marketplace that simplifies browsing, offers organized categorization, and ensures a seamless user interface for furniture selection.
- **Project Goals:**
 - Develop a responsive web application for exploring modern and classic furniture.
 - Provide high-quality visual representation and detailed product information (price, material, category).
 - Implement an efficient search and filtering system for better accessibility.

2. Functional & Non-Functional Requirements

- **Functional Requirements:**
 - **User Authentication:** Secure login system for account access.
 - **Product Discovery:** Browse featured products and categories such as Living Room, Bedroom, and Office.
 - **Search & Filter:** Search by keywords and filter by price range or category.
 - **Shopping Cart:** Ability to add items to a cart for future purchase.
- **Non-Functional Requirements:**
 - **Responsiveness:** UI layout adapts to different screen sizes.
 - **Performance:** Fast loading of high-resolution furniture images.
 - **Reliability:** 24/7 customer support availability and quality guarantee.

3. Software Architecture

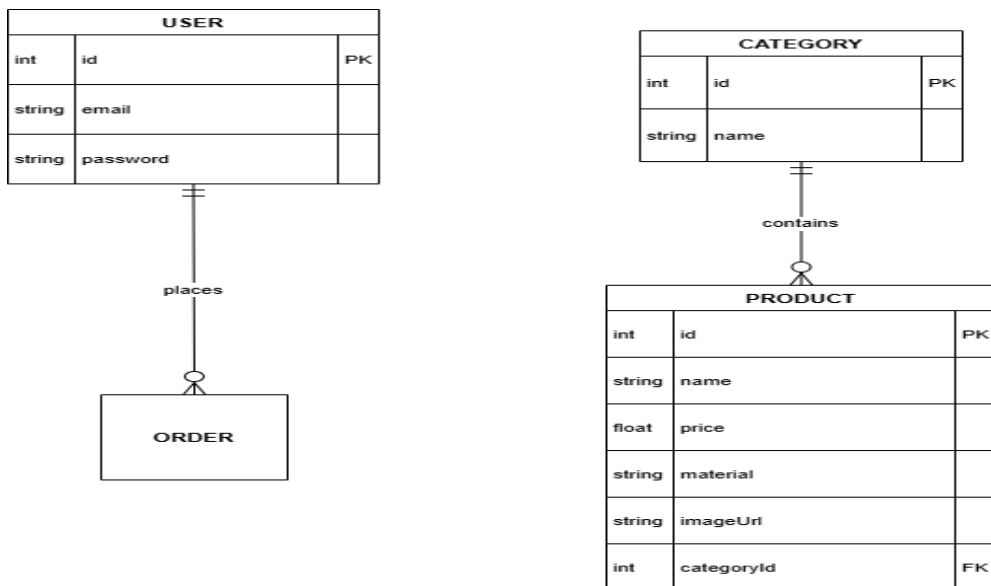
- **Architecture Style: Component-Based Architecture (Angular).**
- **Structure:**

- **Frontend:** Built with **Angular**, utilizing TypeScript for logic and HTML/CSS for structure.
- **Backend:** RESTful API to manage product data, user accounts, and newsletter subscriptions.
- **Database:** Structured schema to store product details, categories, and user credentials.

4. Database Design & Data Modeling

- **Entities:**

- **User:** Email, Password.
- **Product:** Name, Category, Price, Material, Image.
- **Category:** Name (e.g., Living Room, Office, Outdoor).

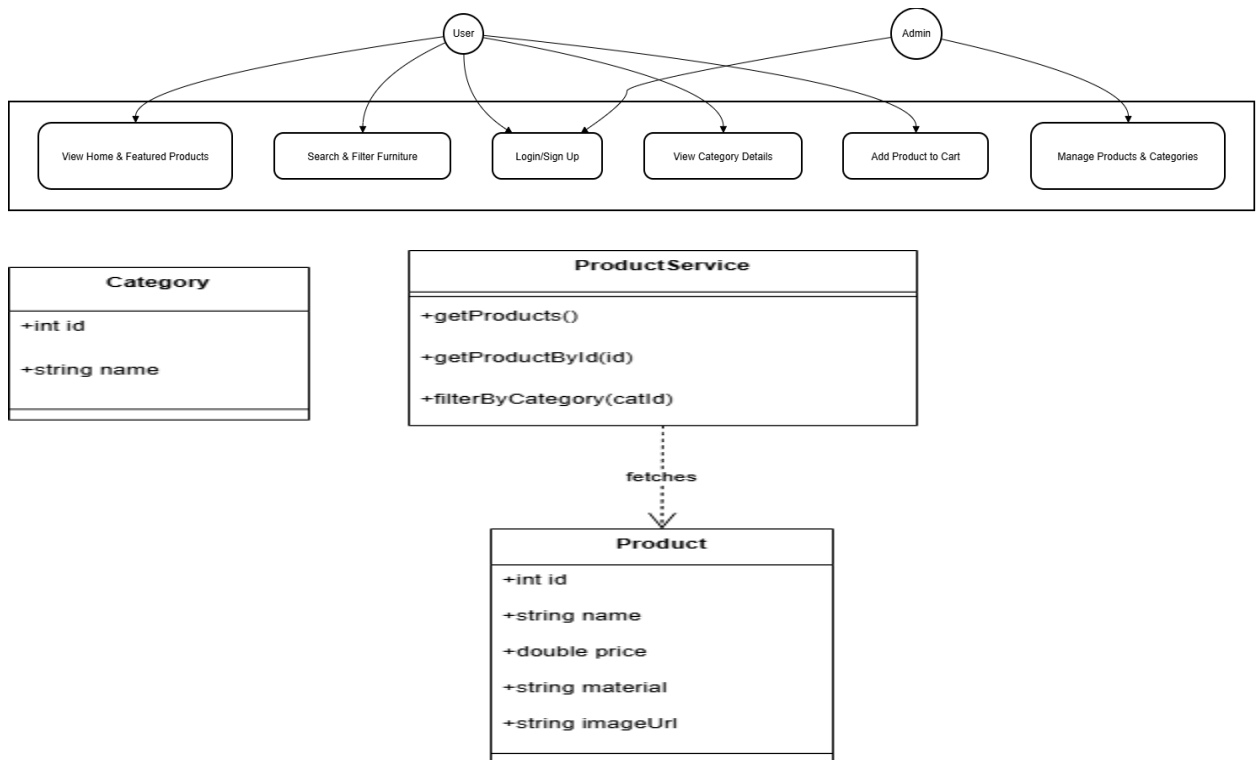


5. Data Flow & System Behavior

- **Process Flow:**

1. User enters the Home Page and views featured items.
2. User filters products by category (e.g., "Dining Room").
3. User selects "Add" to include a product in their selection.

4. System updates the view based on "8 products found" or specific search queries.



6. UI/UX Design & Prototyping

- **Visual Representation:** Clean, minimalist design with a focus on product imagery.
- **Key Screens:**
 - **Hero Section:** High-impact "Transform Your Space" messaging with "Shop Now" call-to-action.
 - **Product Catalog:** Grid layout showing price, name, and "Add" button.
 - **Login Page:** Simple interface for user authentication.
- **UI Guidelines:**
 - **Typography:** Professional sans-serif fonts for readability.
 - **Color Palette:** Neutral tones to emphasize furniture aesthetics.

7. System Deployment & Integration

- **Technology Stack:**
 - **Frontend:** Angular (Framework), TypeScript (Logic), SCSS (Styling).
 - **Communication:** JSON-based API calls for fetching product lists and handling logins.
- **Deployment Strategy:**
 - **Hosting:** The frontend is optimized for modern web browsers.
 - **Integration:** Connection to a newsletter service for "Stay Updated" subscriptions.

8. Testing & Validation

- **Unit Testing:** Testing individual Angular components (e.g., ProductCardComponent).
- **User Acceptance Testing (UAT):** Ensuring the "Apply Filters" and "Search" functions return accurate results as per user expectations.